IMMERSION OF THE STRUCTURE MORPHISMS IS EVENTUAL (CLOSED IMMERSION TRANSITIONS)

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Opens of the limit. Limits, Lemma lemma-descend-opens (same chapter)
- Eventually closed immersion. Limits, Lemma lemma-eventually-closed-immersion (same chapter)

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A0N: sources/stacks-chow/tags.txt, source lines 12641–12641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $X = \lim X_i$ be a directed limit of schemes over S with affine transition morphisms. Assume*

(1) *S quasi-separated,*
(2) *X_i quasi-compact and quasi-separated,*
(3) *the transition morphisms $X_{i'} \to X_i$ are closed immersions,*
(4) *$X_i \to S$ locally of finite type, and*
(5) *$X \to S$ an immersion.*

Then $X_i \to S$ is an immersion for i large enough.

Proof. Choose an open subscheme $U \subset S$ such that $X \to S$ factors as a closed immersion $X \to U$ composed with the inclusion morphism $U \to S$. Since X is quasi-compact, we may shrink U and assume U is quasi-compact. Denote $V_i \subset X_i$ the inverse image of U. Since V_i pulls back to X we see that $V_i = X_i$ for all i large enough by Lemma 01Z4. Thus we may assume $X = \lim X_i$ in the category of schemes over U. Then we see that $X_i \to U$ is a closed immersion for i large enough by Lemma 0A0N. This proves the lemma. □

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